

Noah Blaut

*Enabling situational awareness during
emergency events through the utilization of
inter-building nodes and a mobile application*



Hill Country
CHRISTIAN SCHOOL of Austin



**ENGINEER
YOUR WORLD**
THE UNIVERSITY OF TEXAS AT AUSTIN

The Growing Need

70%

Increase in events

4,000+

Victims

>50%

**in Workplaces and
Schools**



*Federal Bureau of Investigation. FBI Releases 2024 Active Shooter Incidents in the United States Report. 2024.
www.fbi.gov/news/press-releases/fbi-releases-2024-active-shooter-incidents-in-the-united-states-report.*

The Growing Need: Access & Availability

\$10M

**In Security
Improvements
(LISD)**



*Bond 2023. Accessed 16 January 2026. www.leanderisd.org/bond2023/.
Bond: Security Projects. Accessed 16 January 2026.
www.leanderisd.org/bond_securityprojects/.*





The Engineering Design Process

Identify

Establish the
Need

Describe

Identify
Stakeholders
and
Prior
Solutions

Generate

Generate
Concepts and
Brainstorm

Select

Evaluate and
Prototype
Design
Proposals

Embody

Build, Test, &
Refine the
Final Design

Relevant Math/Science

“Timely, accurate, and complete [situational awareness] is a key enabler of successful disaster response when the situation is changing rapidly, resources are limited, and different agencies must coordinate their activities.”

– Kedia et. al.



Relevant Math/Science

Simplex Communication: only one endpoint can send or receive at a time



Relevant Math/Science

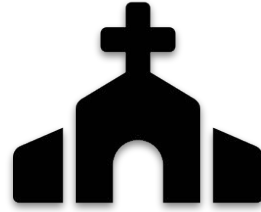
IoT: Internet of Things

IoES: Internet of Emergency Services



Damaševičius, Robertas, et al. "From Sensors to Safety: Internet of Emergency Services (IoES) For Emergency Response and Disaster Management". *Journal of Sensor and Actuator Networks*, vol. 12, no. 3, 2023, Accessed 11 September 2025, p. 41. <https://doi.org/10.3390/jsan12030041>.





Biblical Worldview

"God created mankind in his own image."

Genesis 1:27

"For you formed my inward parts; [. . .] I praise you, for I am fearfully and wonderfully made."

Psalms 139:13–14

**Enabling situational awareness during
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of inter-building nodes and a mobile
application**



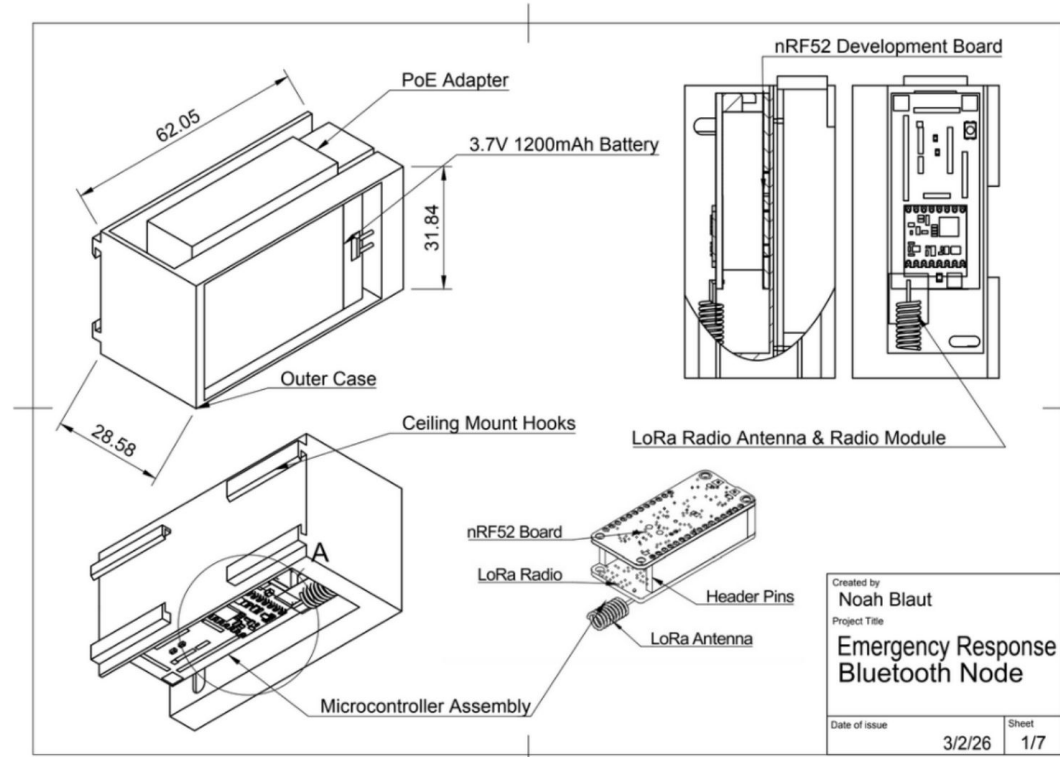
BLE-Based Emergency Localization System



BLE Room Beacons



Mobile Application





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Identify

**~12
min**

Avg. event duration (2000–2013)

**4–6
min**

Recent event response times

28%

Events in workplaces

22%

Events in schools

Tucker, Emma. 'Hard Lockdown': A Timeline of the Apalachee High School Shooting. Sept. 2024, Accessed 11 October 2025. www.cnn.com/2024/09/15/us/apalachee-high-school-shooting-timeline.

Analysis of Existing Solutions

Integrated Emergency Platforms

RapidLite 1516 Log Out

Device Identifier

View:

All Calls XYZ PSAP	
1-655-222-4333 1/23/19 at 4:07PM	1512A
1-588-444-3232 1/23/19 at 3:59PM	1512B
1-984-562-4564 1/23/19 at 3:54PM	1512C
1-456-123-7897 1/23/19 at 3:49PM	1512D
1-789-456-1231 1/23/19 at 3:45PM	1512E

1510

493

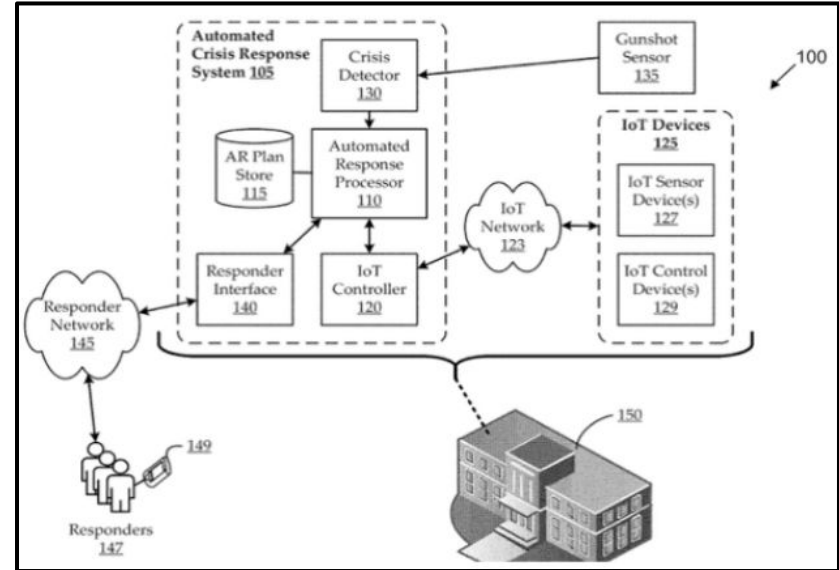
Caller Information 1518A	
Last Updated	Today 10:12AM
Caller Info	
CallerID	15884443232
Caller Name	Jerry Jones
Last Known Location	
Longitude	40.7072690
Latitude	-74.0074330
Uncertainty Radius	5m

Medical Information 1518B	
Medical Info	
Allergies	Nuts, eggs
Birthday	1972-5-1
Blood Type	A+
Disabilities	Mute
Gender	Male
Height	75 inches
Medical Conditions	Heart murmur, Asthma
Medical Notes	Has had open heart surgery



Analysis of Existing Solutions

IoES Automated Crisis Response System





Stakeholder Needs



Interviews with business, local, and state-level emergency response coordinators & personnel

- Direct, delay-free communication between stakeholders
- Rapid and accurate information distribution
- System functional after power or connectivity failures
- Simple interface for untrained personnel
- Integration with existing systems



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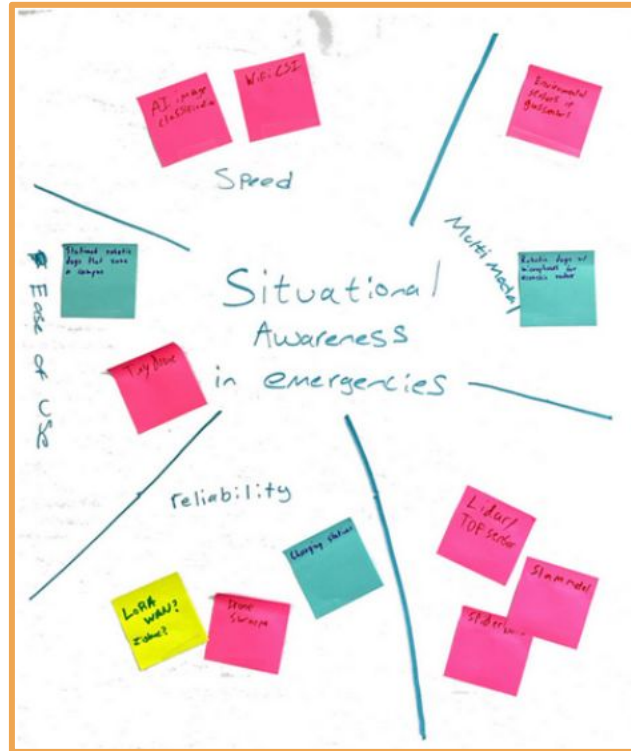
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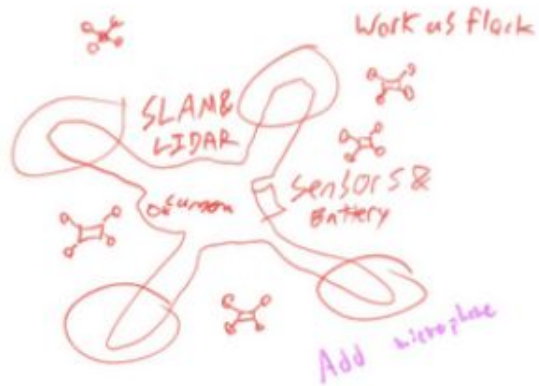
Mind Map





Concept Sketches

Concept A



Concept B





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Pugh Chart #1 – User Interface

Criteria	PTT Radio	VoIP	Smartphone App	Body Pack	Headset
Interoperability	0	+1	-1	-1	-1
Coverage	0	+1	+1	+1	+1
Reliability	0	-1	-1	-1	-1
Latency	0	+1	+1	+1	+1
Duplex	0	+1	+1	+1	-1
Cognitive Load	0	0	+1	-1	-1
Deployment	0	-1	+1	-1	-1
Multimodal	0	0	+1	0	0
Scalability	0	-1	+1	-1	-1
Net Score	0	+1	+5	-3	-4

Better than Datum	Datum	Worse than Datum
+1	0	-1



Pugh Chart #2 – Emergency Coordination

Criteria	Static Guide	Rule-Based	Physical SOS	BLE Localizer	Voice Asst.
Speed	0	0	0	0	-1
Infra. Dependence	0	+1	-1	+1	0
Cost	0	0	-1	-1	-1
Training Burden	0	+1	0	+1	-1
Awareness Quality	0	0	+1	+1	0
Comm. Clarity	0	+1	-1	+1	-1
Hands-Free	0	+1	0	+1	+1
Security	0	+1	-1	+1	0
Net Score	0	+5	-2	+5	-3

Better than Datum	Datum	Worse than Datum
+1	0	-1



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Test #1 – Battery Life

Simulated Infrastructure Degradation (Power Loss)

Node Type	5s Interval	15s Interval	30s Interval
Short Range (BLE)	~ 115 hr	~ 335 hr	~ 950 hr
Long Range (LoRa)	~ 24 hr	~ 80 hr	~ 225 hr

Mean Runtime: +450 Hours (Short Range) | +100 Hours (Long Range)



Test #2 – BLE Localization Accuracy

Received Signal Strength Indicator (RSSI)-to-distance relationship
across power levels and distances

Output Power	Real Distance	RSSI	Calculated Distance
+4 dBm	1 Meter	-42 dBm	1 Meter
	3 Meters	-52 dBm	3 Meters
0 dBm	1 Meter	-44 dBm	1 Meters
	3 Meters	-56 dBm	3 Meters



Test #3 – LoRa Communication Range

Indoor range testing at 900 MHz across building types

Location	Distance	RSSI	Success Rate
Main Building	10 ft	-67 dBm	100%
Main Building	50 ft	-91 dBm	90%
Main Building (Floor 2)	Multi-floor	-103 dBm	64%
Portables	50 ft	-101 dBm	68%

Recommended Density: One Bridge Node per Floor or per Building Cluster





Test #4 – App-to-Mesh Communication

Verifying emergency event propagation and
playbook task assignment

1. Trigger Event



2. Propagate Alert



3. Assign Tasks

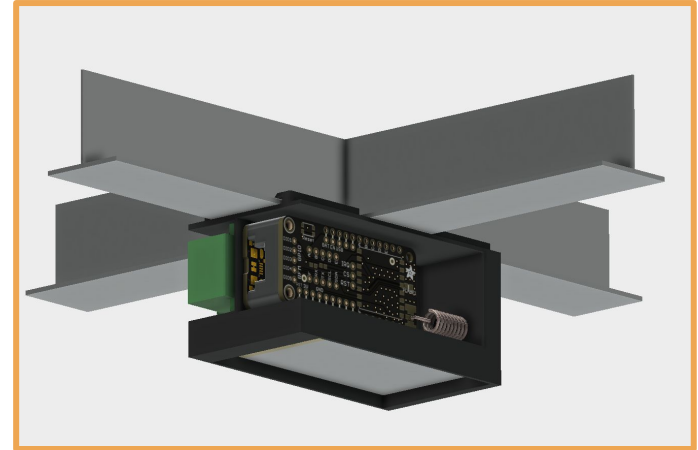
Log analysis confirmed event triggering and task assignment through both the LoRA Mesh & Internet



The Final Design - Room Node

System Components

-  nRF52 System-on-Chip
-  BLE 2.4 GHz Mesh Beacon Broadcasts
-  LoRa 900 MHz Radio
-  3.7V 1200mAh LiPo backup battery
-  Power-over-Ethernet Primary Power

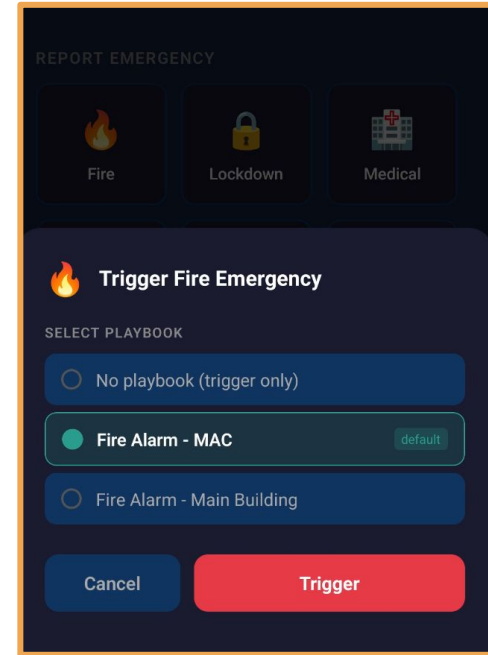


**Long/Short Range Nodes
Tailored to the Environment**



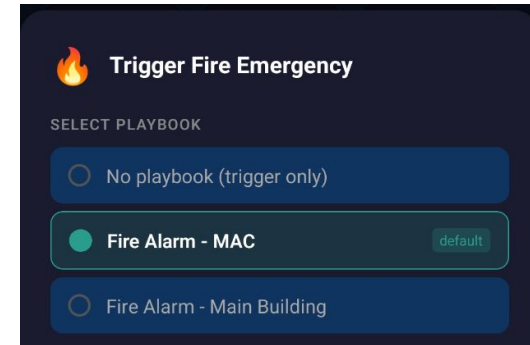
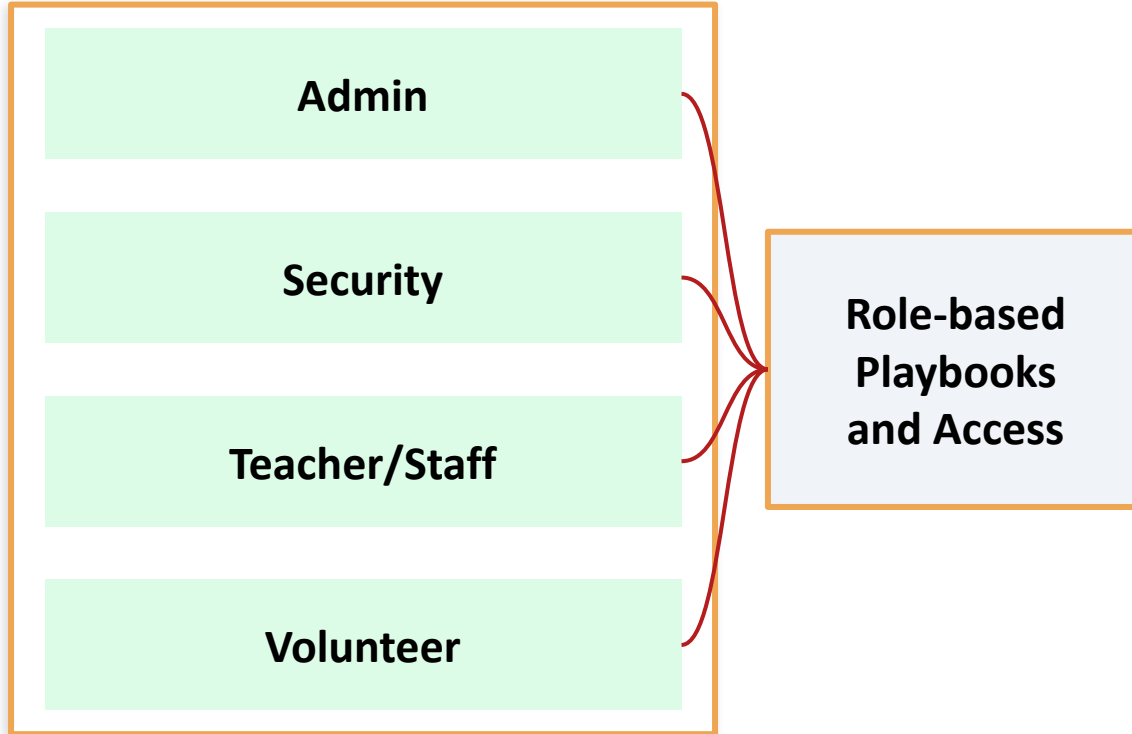
The Final Design - Mobile Application

- 1 BLE beacons broadcast room identifiers continuously
- 2 Mobile app receives signals, calculates user's room via RSSI
- 3 Location displayed on facility floor plan during events
- 4 Emergency alerts sent via internet + mesh network
- 5 Playbook assigns role-specific tasks automatically



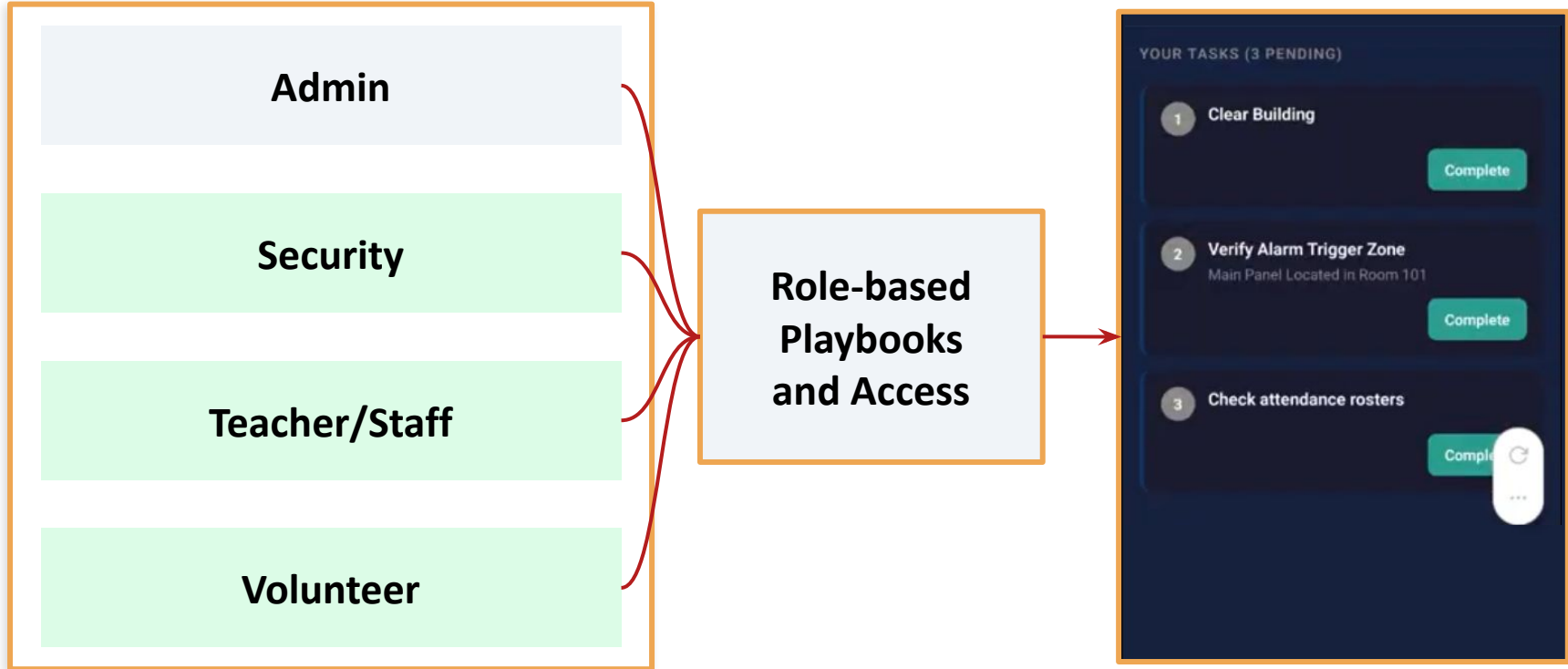


The Final Design - Mobile Application



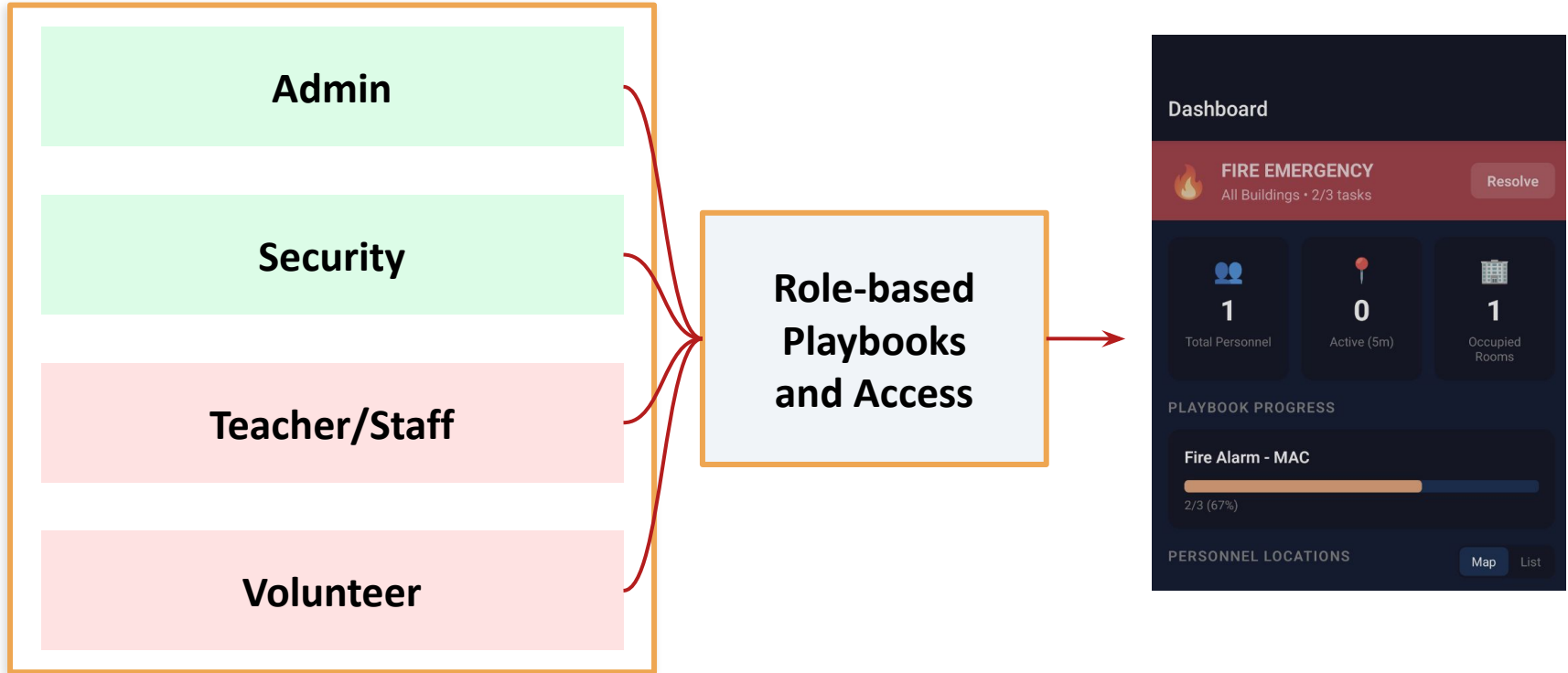


The Final Design - Mobile Application





The Final Design - Mobile Application





Playbook Management Demonstration

←

New Playbook

Description (Optional)

Brief description of this playbook...

Emergency Type


Fire


Lockdown


Medical


Weather


Evacuation


Other

Set as Default

This playbook will be automatically activated for fire emergencies

Creating...

Home

Tasks

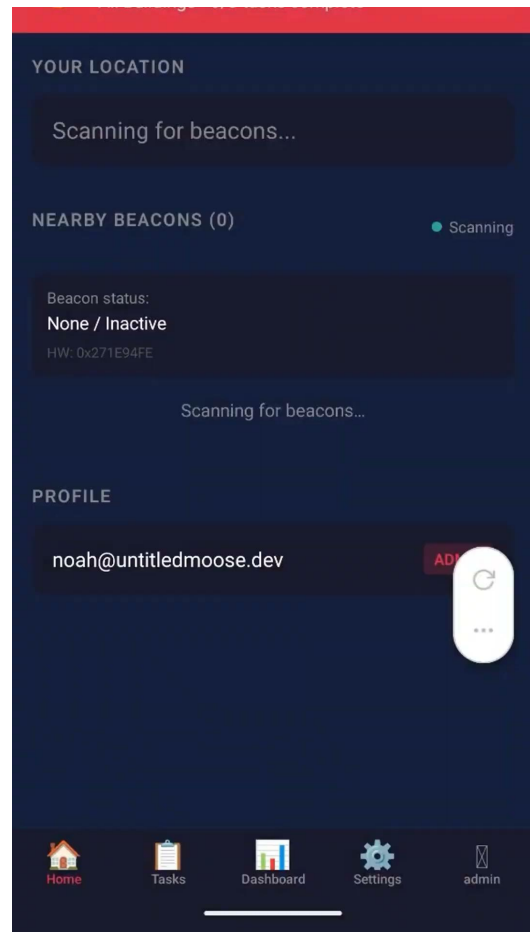
Dashboard

Settings

admin



Playbook Action Demonstration





Design Defense

Counterargument

Reliance on personal mobile devices introduces risk.

Response

Smartphones reduce hardware costs and training burden.



Design Defense

Counterargument

The system relies on power availability during normal operations, so it is not truly "infrastructure-free."

Response

The system has full independent operation during emergencies.



Future Considerations

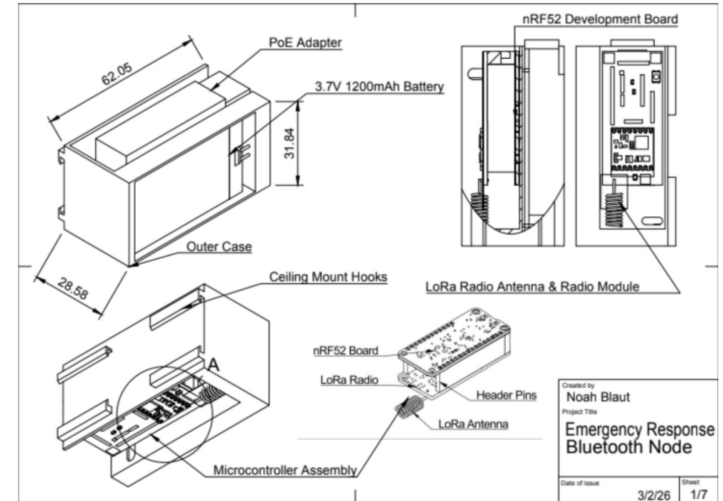
Custom PCB Manufacturing

- Consolidate all components onto a single custom-designed board
- Reduce per-unit cost by eliminating COTS profit margins
- Improve power efficiency using nRF52 ultra-low-power features
- Smaller physical packaging for easier deployment



Conclusion

- Low-cost room-level positioning via BLE beacons
- Mesh-based communication independent of building infrastructure
- Role-specific playbooks reduce cognitive load
- Lowers the barrier for schools, churches, and organizations





Thank You

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